LAB 7: Building an Interactive Dynamic Dashboard with JavaScript

Lab Objective

In this lab, you will apply the core JavaScript concepts introduced in Lecture 7 to build an interactive, single-page **System Health & Energy Monitor**.

Working without starter code templates, you will construct the HTML structure, style it using Bootstrap 5 classes, and write custom JavaScript to handle user events, modify DOM text and classes, and execute stateful calculations.

Project Requirements & Open-Ended Scenario

You have been hired to create a prototype monitoring panel for a local facility. The panel needs to track two interactive systems:

1. **System Power Switcher (Status Monitor):** An interactive panel that allows users to toggle system power between **ONLINE**, **STANDBY**, and **CRITICAL** states.
2. **Energy Load Calculator (Stateful Counter):** A load-testing tool that increases or decreases system energy output in megawatt (MW) increments, providing real-time visual feedback based on the total load value.

Rubric Breakdown (100 Points Total)

Your lab submission will be evaluated against the following itemized criteria:

Criteria	Description	Points
1. File Setup & Script Linkage	HTML boilerplate linked to Bootstrap 5 CDN via <head>, containing a single <script> block placed right before the closing </body> tag. Zero console errors on page load.	**15 pts**
2. Variable Declarations & Scope	Correct, explicit use of modern ES6 keywords (const for DOM element selections and fixed values; let for mutable numerical load state). Zero usage of legacy var.	**15 pts**

Criteria	Description	Points
3. System Power Switcher (DOM & Classes)	Three functional state buttons (Online, Standby, Critical). Clicking each updates the status text badge and dynamically updates Bootstrap contextual background classes (bg-success, bg-warning, bg-danger).	**25 pts**
4. Energy Load Calculator (Stateful Logic)	Functional +10 MW and -10 MW controls with a reset option. State variable correctly updates without dropping below 0 MW. Load text and progress bar visually adapt based on load thresholds.	**25 pts**
5. Code Structure, Formatting & Comments	Clean, readable code structure with clear function definitions, consistent indentation, meaningful variable names, and clear comments explaining key DOM operations.	**20 pts**

Detailed Implementation Guide

Part 1: Page Layout & HTML Structure (index.html)

Build a single-page HTML dashboard structured with Bootstrap 5 container grid classes.

1. **Header Section:** A title header displaying "System Health & Energy Monitor".
2. **Card 1: System Power Switcher:**
    - A status badge displaying the current state (Default: ONLINE).
    - Three control buttons: **Online** (Green/Success), **Standby** (Yellow/Warning), and **Critical** (Red/Danger).
3. **Card 2: Energy Load Calculator:**
    - A display showing current load in **Megawatts (MW)** (Default: 0 MW).
    - A dynamic Bootstrap progress bar representing the current load percentage (max 100 MW).

- Three action buttons: **+10 MW**, **-10 MW**, and **Reset Load**.

Part 2: JavaScript Implementation Instructions

Inside your <script> tag at the bottom of index.html, implement the following functional requirements:

A. DOM Element Selection

- Select all required buttons, badges, load text displays, and progress bar elements using document.getElementById().
- Store references in const variables.

B. System Power Switcher Logic

- Define a function or event listeners for the status buttons.
- When a button is clicked:
  - Update the inner text of the status badge.
  - Dynamically remove existing color classes and apply the class corresponding to the active status (bg-success, bg-warning text-dark, bg-danger).

C. Energy Load Calculator Logic

- Declare a state variable using let currentLoad = 0;.
- Write logic for increasing load by 10 (up to a max of 100 MW) and decreasing load by 10 (minimum 0 MW).
- Update the load text display and progress bar width dynamically (element.style.width = currentLoad + "%").
- **Threshold Styling Rule:**
  - 0 - 49 MW: Progress bar class bg-info
  - 50 - 79 MW: Progress bar class bg-warning
  - 80 - 100 MW: Progress bar class bg-danger

Submission Instructions

1. Save your completed code as an index.html file in your project directory.

2. Open index.html in your web browser and test all buttons, states, and load boundaries.
3. Open Developer Tools (F12 or Right-Click -> Inspect -> Console) to verify that there are zero syntax or reference errors on load and during interactions.